WORKED SOLUTION

STEP I 2001, Question 14

Probability and Statistics • Random Orderings

STEP I 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q14

This question has a full guided treatment in *Beyond STEP*, Section 12.12, *When to Stop Looking*: https://beyondstep.keithmcnulty.org/ch12/secretary.html

The key idea

Don't think about the whole interview order: think only about *where particular candidates sit*. The best of the first n is the best overall exactly when the top candidate is among the first n interviewed, and by symmetry the top candidate is equally likely to be in any of the N positions. Part (ii) then needs one extra case, or a neat complement.

Solution

Label the candidates by rank, so candidate 1 is the best overall, candidate 2 the second best, and so on. Since all $N!$ orders are equally likely, every candidate is equally likely to occupy any given position, and the set of candidates seen in the first n interviews is equally likely to be any of the $\binom{N}{n}$ subsets of size n. Throughout, $1 \leq n \leq N$.

Part (i)

Let A be the event that candidate 1 is among the first n interviewed. If A happens, candidate 1 is certainly the best of the first n; if it doesn't, the best of the first n is someone worse than candidate 1. So the event we want is exactly A.

Candidate 1 is equally likely to be in each of the N positions, and n of these are among the first n, so

$$\mathrm{P}(\text{best of first } n \text{ is best overall}) = \frac{n}{N}.$$

Part (ii)

Now take $N \geq 2$ so that there is a second-best candidate. The best of the first n is the best or second best overall if and only if

- candidate 1 is among the first n (event A), or
- candidate 1 is *not* among the first n, but candidate 2 is.

> **Ask yourself** In the second case, how do we know the best of the first n is candidate 2, and not someone even better?

The only candidate better than candidate 2 is candidate 1, who in this case was not interviewed early. Conversely, if neither 1 nor 2 is among the first n, the best of the first n is ranked 3 or worse. So the two cases above are exactly the event we want, and they are mutually exclusive.

The first case has probability $\frac{n}{N}$ from part (i). For the second, $\mathrm{P}(A') = \frac{N-n}{N}$, and given that candidate 1 is in one of the last $N-n$ positions, candidate 2 is equally likely to be in any of the remaining $N-1$ positions, n of which are among the first n. Hence

$$\begin{aligned}\mathrm{P}(\text{best of first } n \text{ is first or second overall}) &= \frac{n}{N} + \frac{N-n}{N} \cdot \frac{n}{N-1} \\ &= \frac{n\big[(N-1)+(N-n)\big]}{N(N-1)} \\ &= \frac{n(2N-n-1)}{N(N-1)}.\end{aligned}$$

Alternatively, use the complement: the event fails exactly when all n early candidates come from the $N-2$ candidates ranked 3 to N, which has probability

$$\frac{\binom{N-2}{n}}{\binom{N}{n}} = \frac{(N-n)(N-n-1)}{N(N-1)},$$

and $N(N-1) - (N-n)(N-n-1) = 2Nn - n^2 - n = n(2N-n-1)$, giving the same answer.

The case $N = 4$, $n = 2$

The formulas give $\frac{2}{4} = \frac{1}{2}$ for (i) and $\frac{2 \cdot 5}{4 \cdot 3} = \frac{5}{6}$ for (ii).

To list the possibilities, note that only *which* two candidates are interviewed first matters. Each of the $\binom{4}{2} = 6$ pairs arises from exactly $2! \times 2! = 4$ of the 24 orders (order the first two, then the last two), so the six pairs are equally likely:

First two interviewed	$\{1,2\}$	$\{1,3\}$	$\{1,4\}$	$\{2,3\}$	$\{2,4\}$	$\{3,4\}$
Best of these	1	1	1	2	2	3

The best of the first two is the best overall in 3 of the 6 cases, probability $\frac{1}{2}$, and is the best or second best in 5 of the 6 cases, probability $\frac{5}{6}$. Both agree with the formulas.

Checks

- $n = N$: we see everyone, so both probabilities should be 1. Indeed $\frac{N}{N} = 1$ and $\frac{N(N-1)}{N(N-1)} = 1$.
- $n = 1$: the first candidate is a uniformly random one, so (ii) should be $\frac{2}{N}$; the formula gives $\frac{2N-2}{N(N-1)} = \frac{2}{N}$.
- The answer to (ii) is at least the answer to (i), since $2N - n - 1 \geq N - 1$.
- A full enumeration of all $N!$ orders for $N \leq 7$ and every n agrees exactly with both formulas (for example $N = 5$, $n = 2$ gives $\frac{2}{5}$ and $\frac{7}{10}$).

Common slips

- Forgetting in (ii) that "second best overall" only counts when the best overall is *not* among the first n; simply adding $\frac{n}{N} + \frac{n}{N}$ double counts.
- Using $\frac{n}{N}$ for candidate 2 after conditioning on candidate 1's position: once candidate 1 is placed, only $N - 1$ positions remain.

Beyond STEP

This is the setting of the famous *secretary problem*: if you must accept or reject each candidate on the spot, the best strategy is to watch the first n and then accept the next candidate better than all of them, and an extension of the argument in part (i) shows that the optimal n is roughly N/e, succeeding with probability about $1/e$. The symmetry argument used here ("each candidate is equally likely to be anywhere") is the starting point for the study of random permutations. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of symmetry arguments like these.

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